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Energy Management of Microgrids: An MPC-Based Techno-Economic Optimisation for RES Integration and ESS Utilisation

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ABSTRACT

The rapid growth in electricity demand poses significant challenges to power systems that require efficient energy management frameworks. This paper proposes a real-time energy management framework for DC residential microgrids based on model predictive control (MPC) and a mixed integer nonlinear programming (MINLP) optimisation approach. Unlike conventional methods relying on static datasets, the proposed framework integrates real-time optimisation and predictive data derived from microgrid simulations, which enhances adaptability, scalability and operational efficiency. A demand response (DR) programme for residential loads further reduces reliance on the upstream grid during peak hours. The performance of the framework is validated across multiple scenarios, like increased demand and fluctuations in renewable energy and demand. Results indicate significant economic benefits through optimised power exchange with the upstream grid, achieving cost reductions of 21.05%, 18.36% and 16.59% under real-time, three-tariff and fixed tariffs, respectively, compared to conventional MPC methods. Moreover, it minimises state of charge (SoC) changes to a maximum of 17.4%, significantly reducing energy storage depreciation and extending longevity as technical performance. These findings demonstrate the framework's ability to adapt dynamically to operational changes, optimise power exchanges and achieve technical and economic goals, making it a scalable solution for real-time residential microgrid energy management.

1 | Introduction

1.1 | Motivation

It is predicted that 70% of the world population will live in cities by 2050 and the residential sector in cities will account for 40% of total electricity consumption [1]. Simultaneously, the average annual growth rate of 4.3% in global electricity consumption places an ever-increasing pressure on power systems worldwide [2]. This rising demand, particularly in the residential sector, underscores the need for cost-effective and efficient energy

management solutions. Residential microgrids have emerged as a promising approach, which offers localised energy management while facilitating the integration of renewable energy sources (RESs) and energy storage system (ESS) utilisation to enhance resilience, reliability and decarbonisation [3, 4]. However, achieving these benefits requires the development of an advanced energy management system (EMS) that can optimise microgrid operation based on technical and economic goals. Despite progress in EMS, current methods face significant limitations in adapting to real-time dynamics and handling operational complexities.

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Heuristic approaches, such as fuzzy logic control (FLC) and artificial neural networks (ANNs), although widely employed, are inherently constrained by their reliance on static data and predefined parameters. These methods often limit the capture of the stochastic and dynamic nature of microgrids, resulting in suboptimal performance, particularly in the presence of fluctuating RES or demands [5, 6]. FLC-based models primarily focus on single energy sources, such as plug-in hybrid electric vehicles (PHEVs), using fuzzy inference systems (FIS) for state of charge (SoC) estimation. These models lack flexibility when dealing with RESs or islanded operations [7, 8]. Similarly, ANN-based methods, while effective in renewable energy forecasting using tools like numerical weather prediction (NWP), struggle with real-time adaptability, particularly in scenarios involving electricity price fluctuations or unpredictable demand changes [9, 10].

In addition to the mentioned heuristic methods, advanced control techniques, particularly model predictive control (MPC), have shown considerable promise for optimal EMS. MPC is selected as the control strategy in this paper due to its ability to predict future status, dynamically optimise power flow by system behaviour prediction over a time horizon, minimise an objective function based on both technical and economic constraints and adapt to real-time variations in system conditions [11, 12]. Also, in microgrid energy management, MPC is commonly used at the tertiary control level, which focuses on optimal energy management (whereas primary and secondary control levels focus on voltage-frequency regulation and power quality control, respectively) [13]. However, MPC methods often depend on predefined datasets, which limits their ability to incorporate instantaneous measurements to adapt in real-time to uncertainties in RES and demand and dynamic electricity prices.

The growing reliance on microgrids to deal with global energy challenges, coupled with the inadequacies of existing methods, underscores the need for innovative, real-time adapted solutions to operational changes. This paper addresses these needs by introducing an MPC-based EMS framework that integrates predictive data based on the performance of microgrids and instantaneous data for real-time adaptability. By real-time optimisation based on both technical and economic performance, the proposed framework provides an efficient and sustainable energy management solution to maximise the potential of residential microgrids.

1.2 | Literature Review

The optimisation of energy management in microgrids has gained considerable attention due to the growing demands for energy efficiency and the seamless integration of RESs and ESSs into microgrid operations. Hybrid power systems that incorporate photovoltaic panels (PV), wind turbines (WT), fuel cells (FC) and ESSs offer a sustainable solution to these challenges [14, 15]. Among the various methodologies developed, MPC has emerged as a robust framework to optimise key metrics like cost of energy (COE) and net present cost (NPC). So, MPC frameworks utilise cost functions to allocate power resources and manage ESS operations optimally, often leveraging meteorological data [16, 17]. However, conventional MPC methods often rely on static inputs, such as predefined data and ignore real-time operational

constraints related to ESS depreciation and demand fluctuations. These limitations reduce their effectiveness in dynamic and unpredictable environments, leading to suboptimal results under real-world conditions.

To address uncertainties in RES and demand, advanced MPC frameworks such as Robust MPC and Stochastic MPC have been proposed. Robust MPC incorporates forecast errors and operational constraints, effectively balancing technical and economic efficiency, which have demonstrated success in reducing costs for islanded microgrids [18, 19]. However, their limited adaptability to real-time variations remains a significant challenge for microgrids operating under unpredictable operational changes. Stochastic MPC, on the other hand, employs scenario-based optimisation to manage forecast errors and ensure optimal operation under uncertainties [20]. This method has shown promise in enhancing off-grid microgrid performance by accounting for short-term forecast errors as disturbances during decision-making processes [21, 22]. Multi-layer MPC-based approaches have also been developed to optimise power exchange and ESS utilisation [23, 24]. These methods employ a two-layer structure: the first layer determines the power purchased from the upstream grid and the second layer optimises ESS operations. While these approaches improve energy management, their computational complexity often limits scalability in large-scale microgrids.

Economic MPC frameworks have been explored for their potential to minimise operational costs, particularly for microgrids equipped with electric vehicles (EVs). Hierarchical economic MPC approaches have been employed for managing EV charging in microgrids, where an EMS operation aggregator determines the optimal placement of charging stations and allocates power dynamically [25, 26]. Hybrid methods, such as combining particle swarm optimisation (PSO) with economic MPC have achieved substantial cost reductions but face challenges in handling complex constraints and long-term impacts [27]. Similarly, genetic algorithms (GA) have been integrated with MPC to improve resource allocation and adapt to changing operational conditions [28–30]. While these methods enable the optimisation of multi-objective functions, they often ignore ESS depreciation or the variability of dynamic electricity pricing comprehensively. However, economic MPC strategies applied to PV-diesel generator systems have demonstrated cost reductions of up to 15% compared to conventional methods [31–33]. So, these cost-effective frameworks frequently overlook ESS longevity and its associated economic implications.

Distributed MPC has emerged as an effective method to improve energy management in microgrids, which enables decentralised optimisation while preserving the privacy of individual microgrids. Distributed MPC frameworks optimise power flows through coordinated operations, using proportional dual decomposition to achieve cost-effective performance without compromising system stability in microgrids [34–36]. Additionally, peer-to-peer power trading frameworks within residential microgrids have demonstrated the potential to optimise surplus power sharing among interconnected buildings to reduce costs [37]. Demand response (DR) models managed by energy aggregators have also shown promise under uncertainties in system performance by dynamically allocating resources during critical periods [38]. In this DR, a two-level MPC model integrates

the reinforcement learning model to handle different time-scale controllers.

Despite these significant advances, there are critical limitations in existing EMS microgrid approaches that highlight the need for further innovation. One of the challenges is the insufficient real-time adaptability of many frameworks. Numerous studies rely on day-ahead scheduling methods that are unable to respond effectively to dynamic changes in RES generation or fluctuating demand [39, 40]. This reliance results in reduced responsiveness and efficiency in energy management systems, particularly under rapidly changing operational conditions [41–43]. Innovative strategies such as advanced metering infrastructure (AMI) and demand-side management frameworks like vehicle-to-campus (V2C) [44] and bicycle-to-campus (B2C) [45] have been introduced to address peak demand issues [46]. While these approaches alleviate stress on power systems during critical periods, their lack of integration with real-time optimisation and predictive control severely limits their capacity to balance supply and demand dynamically when faced with unexpected variations in RES generation or consumption [47].

Another significant limitation is the inadequate consideration of ESS depreciation in existing studies and only a few MPC-based strategies overlook the long-term effects of SoC changes on storage systems. Frequent SoC variations often lead to reduced ESS longevity, increased replacement costs and affect the long-term economy of microgrids [48, 49]. Furthermore, economic uncertainties such as dynamic pricing and demand fluctuations are often neglected in existing methods [50–52]. This oversight prevents microgrids from maximising economic benefits while maintaining technical objectives, such as minimising ESS depreciation. Addressing these challenges requires advanced EMS frameworks that integrate real-time adaptability, explicitly model ESS longevity and incorporate dynamic pricing considerations. Such innovations would enable microgrids to enhance both technical performance and long-term economic sustainability. A comparison of previous studies is presented in Table 1, which summarises their key characteristics and limitations in microgrid energy management.

This table is derived from a comprehensive review of the existing literature, highlighting critical aspects such as economic optimisation, tariff handling for dynamic electricity pricing and the integration of RES and ESS depreciation into the technical goals of energy management. Incorporating bidirectional grid-connected microgrid capabilities enables bidirectional power exchange between the microgrid and the upstream power grid. This comparative analysis reveals that while many frameworks achieve partial optimisation, they often fail to incorporate real-time adaptability or performance-based prediction to dynamically adjust to operational changes instead of using predetermined data. The proposed framework addresses these gaps, offering an integrated solution to optimise economic and technical objectives in real-time.

1.3 | Research Gaps

Despite the progress in microgrid energy management systems, several critical research gaps remain, as identified in Table 1.

Addressing these gaps is vital for achieving scalable, cost-effective and sustainable microgrid operations. The primary research challenges are summarised as follows:

- i. **Economic optimisation based on tariff systems with ESS depreciation considerations:** Many existing studies prioritise economic optimisation but ignore incorporating dynamic tariff systems, such as real-time or three-tariff pricing. This oversight limits the potential for achieving substantial cost savings. Moreover, while economic models are essential, they should integrate technical constraints like ESS depreciation. Ignoring factors like the SoC dynamics often results in extreme SoC level changes (SoC remains near 100% or 20%), significantly increasing ESS depreciation. An effective EMS should simultaneously optimise costs and ESS longevity by maintaining SoC within safe operational ranges and economic optimisation performances.
- ii. **Real-time adaptability and uncertainty management:** Microgrids inherently face uncertainties due to the variable nature of RES and dynamic demand fluctuations. However, some MPC frameworks neglect to utilise real-time optimisation capabilities for adaptive decision-making, leading to sub-optimal performance during unexpected changes in RES generation or demand. MPC provides a dynamic control approach, which optimally predicts and adjusts system performance by incorporating real-time data and future uncertainties. So, an MPC-based EMS should use its predictive and feedback control capabilities to dynamically adjust power flows and ESS operations to adapt to reliable and cost-effective performance while effectively managing real-time uncertainties.
- iii. **Dependence on predefined data in some MPC frameworks:** Conventional MPC-based methods heavily rely on static, predefined datasets, which restrict their applicability in dynamic real-world scenarios. This dependence reduces their adaptability to respond to rapidly evolving operational conditions. The proposed framework addresses this limitation by integrating instantaneous measurements with predictive data derived from microgrid simulations, enabling performance-based predictions that dynamically adapt to operational changes. This integration enhances both the scalability of the EMS for large-scale microgrids and practical applicability.
- iv. **Bidirectional power flow with the upstream grid:** Bidirectional power exchange with the upstream grid is a critical feature for economic and technical optimisation, yet it is often overlooked. The ability to purchase power during low-price periods and sell surplus power during high-price periods is essential for economic viability. Additionally, microgrids operating in an islanded mode without a two-way connection risk load losses during periods of high demand or reduced RES generation. An effective EMS must facilitate seamless bidirectional power flows to mitigate these risks and optimise economic and technical performance.

1.4 | Contributions

This study addresses the identified gaps by presenting a real-time EMS optimisation framework for a residential micro-

TABLE 1 | Comparative analysis of key characteristics in previous studies.

Ref. No	Economical optimi- sation	Real-time adapt- ability	Tariff	Components			ESS depreciation	Bidirectional grid- connected	Performance- based prediction	Optimi- sation method
				PV	WT	ESS				
[14]	✓	×	✓	✓	✓	✓	×	✓	×	Convex
[15]	×	✓	×	✓	×	✓	✓	×	✓	Quadratic programming
[16]	✓	✓	×	✓	✓	✓	×	×	×	PSO
[17]	✓	✓	✓	✓	✓	✓	×	✓	×	MILP
[18]	✓	✓	×	✓	✓	✓	×	✓	×	Robust linear
[19, 31]	✓	✓	×	✓	✓	✓	×	×	✓	Robust
[20]	✓	✓	×	✓	✓	✓	×	✓	×	Stochastic
[21]	×	✓	×	✓	✓	✓	✓	×	✓	MILP
[22]	✓	×	✓	✓	×	✓	×	×	✓	Stochastic
[23]	✓	×	×	✓	✓	✓	×	✓	×	MINLP
[24]	×	✓	×	×	✓	✓	✓	×	✓	PSO
[25, 26]	✓	✓	×	✓	×	✓	×	✓	×	Pareto
[27, 30, 32]	✓	×	×	✓	×	✓	✓	✓	×	PSO
[28]	✓	✓	×	✓	✓	✓	×	×	×	GA
[29]	✓	×	×	✓	✓	×	×	✓	×	PSO
[33]	✓	×	✓	✓	×	×	×	✓	×	Linear programming
[34]	✓	×	✓	✓	✓	✓	✓	×	×	Robust
[35]	×	✓	×	✓	×	✓	✓	✓	×	MIQP
[36]	✓	×	✓	✓	✓	×	×	✓	×	Distributed
[37, 43]	✓	×	✓	✓	×	✓	×	✓	×	Non-linear programming
[38]	✓	✓	×	✓	×	✓	×	✓	×	MILP
[39]	×	×	×	✓	✓	✓	✓	✓	×	Coordinated
[40]	✓	✓	✓	✓	✓	✓	×	×	×	MILP
[41]	✓	✓	×	✓	✓	✓	×	✓	×	Distributed
[42]	✓	✓	×	✓	✓	✓	×	✓	×	Peer-to-peer
[45]	✓	×	×	✓	✓	✓	×	✓	×	Quasi- Newton
[46]	✓	×	×	✓	×	✓	×	✓	×	Linear programming
[47]	✓	×	✓	✓	×	✓	×	✓	×	MILP
[48]	✓	×	×	✓	✓	✓	✓	✓	×	MILP
[49]	✓	✓	×	✓	✓	✓	×	✓	×	Stochastic
[50]	✓	×	×	✓	×	✓	×	✓	×	Quadratic
[51]	✓	×	✓	✓	✓	✓	×	×	×	MILP
This Paper	✓	✓	✓	✓	✓	✓	✓	✓	✓	MINLP

grid, designed to enhance both technical and economic performance. The framework integrates MPC with MINLP to address the limitations of key gaps identified in the literature, as shown in Table 1. The use of MINLP overcomes the shortcomings of conventional optimisation methods, such as convex optimisation, PSO and MILP. Convex optimisation

often oversimplifies problems, leading to suboptimal solutions, while metaheuristics like PSO struggle with inefficiencies in large-scale systems. Similarly, MILP is inadequate for modelling nonlinear equations in real-world microgrid constraints. The proposed MINLP approach effectively combines discrete decision-making with accurate modelling of nonlinear dynamics

to ensure robust optimisation for complex microgrid operations.

The framework incorporates real-time performance-based predictive data from microgrid simulations over a 24 h horizon. Unlike conventional MPC methods that rely on static datasets, the proposed approach dynamically updates predictions with instantaneous measurements, enabling adaptive and practical decision-making under changing conditions. The proposed objective function integrates the individual cost functions of ESS, RES, the upstream grid and residential loads, considering technical and economic constraints such as power flow limitations and SoC boundaries. This comprehensive approach ensures optimal economic and technical performance, which achieves all characteristics outlined in Table 1. By optimising SoC changes and exchanging power between the microgrid and the upstream grid, the framework improves ESS longevity and economic savings as technical and economic optimisation, respectively. A DR program is incorporated to curtail non-essential loads during peak demand periods, which reduces reliance on upstream power systems and enhances economic savings.

The proposed residential DC microgrid comprises four components: ESS including lithium-ion and lead-acid batteries, RESs consisting of photovoltaic panels and wind turbines, residential loads and an upstream power distribution grid. Power flow within the microgrid is managed through bidirectional AC/DC inverters for selling and purchasing power, and DC/DC converters facilitate ESS charge/discharge operations. Additionally, the upstream grid is modelled by transformers and transmission line parameters to enhance the realism of the simulation. The proposed EMS framework achieves the following objectives, which demonstrate the superiority of the proposed framework over the conventional MPC method, and each corresponds to the research gaps mentioned in the previous subsection and covers them:

- i. Techno-economic optimisation simultaneously: In addition to optimising the power flow like similar methods, the proposed method simultaneously optimises the technical and economic performance of the microgrid, considering electricity price based on tariff systems while observing the operational constraints of the microgrid components. Optimising power exchanges with the upstream grid based on dynamic pricing signals achieves significant cost reductions by efficiently utilising RES and ESS and minimises grid reliance during peak-price periods.
- ii. Effective management of demand, RES and electricity price fluctuations: Using real-time performance-based prediction allows the framework to react faster and more accurately to sudden changes in demand, RESs and electricity prices. It demonstrates the effective management of the proposed method to ensure supply and demand balance with optimal utilisation of RESs, ESS and the upstream grid.
- iii. Real-time and adaptive performance: Since the proposed framework is based on MPC, the optimisation process is also performed using MINLP and both are designed for real-time execution, the proposed method ensures real-time performance, making it applicable for operation in different conditions and large-scale microgrids. Also, using real-time performance-based prediction data from microgrid simula-

tions at each time step allows any operational changes to be taken into account in the proposed energy management system, which improves its adaptability.

- iv. Bidirectional exchanged Power optimisation and SoC management to extend ESS longevity: The proposed framework minimises SoC variation to reduce ESS degradation and extend battery longevity. To achieve this, power and SoC constraints are incorporated into the optimisation model and by solving the objective functions in real-time, optimal SoC values are determined to ensure efficient charge/discharge operations. Furthermore, the bidirectional power exchange ability enables the energy management system to sell excess power to the grid, thereby increasing economic benefits.

Overall, this research contributes a scalable, real-time, adaptive and sustainable solution for energy management in microgrids by integrating predictive analytics based on the performance of microgrids, real-time adaptability and economic and technical optimisation using MINLP techniques. The main contributions of this study can be summarised as follows:

- Optimising energy resource usage and grid interactions based on dynamic tariff systems to maximise economic benefits.
- Reducing SoC changes to minimise ESS depreciation and ensure sustainable operation.
- Incorporating ESS charge/discharge constraints, RES variability, grid power flows and performance-based operation prediction to enhance the adaptability of the framework.
- Introducing load curtailment strategies during peak periods to improve economic and technical performance.
- Providing a scalable framework suitable for residential microgrids, with potential extensions to larger systems.

The remainder of this paper is organised as follows for presenting this framework: The modelling of the microgrid and the objective functions are presented in Section 2. The EMS architecture and implementation are described in Section 3. Section 4 includes the simulation results for various scenarios, which shows the effectiveness of the framework. Finally, the conclusions and suggestions for further research directions are expressed in Section 5.

2 | Modelling and Formulation of Microgrid

A detailed residential microgrid, shown conceptually in Figure 1, is modeled to evaluate the proposed framework. This microgrid integrates PV panels, WT, ESS (lithium-ion and lead-acid batteries), residential loads and the upstream grid. All components are connected to a Point of Common Coupling (PCC), managed through converters and inverters to ensure seamless power flow:

- ESS: ESS is connected to the PCC through a bidirectional DC/DC converter, which enables controlled charging and discharging of the batteries (Figure 2a).
- Electrical loads: Loads are supplied from the PCC using a AC/DC inverter (Figure 2b).

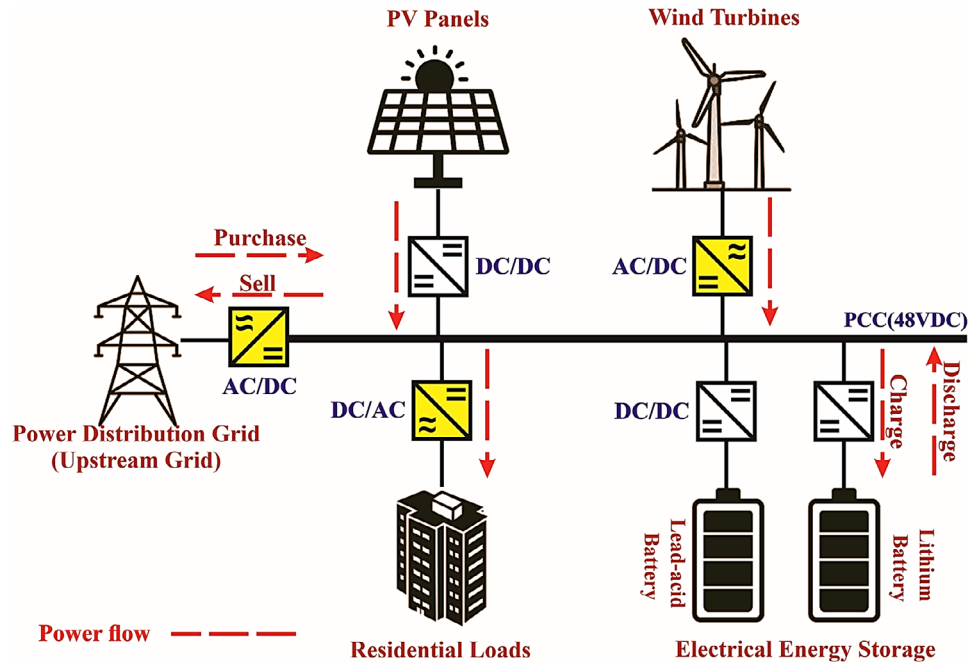


FIGURE 1 | The conceptual model of the microgrid components.

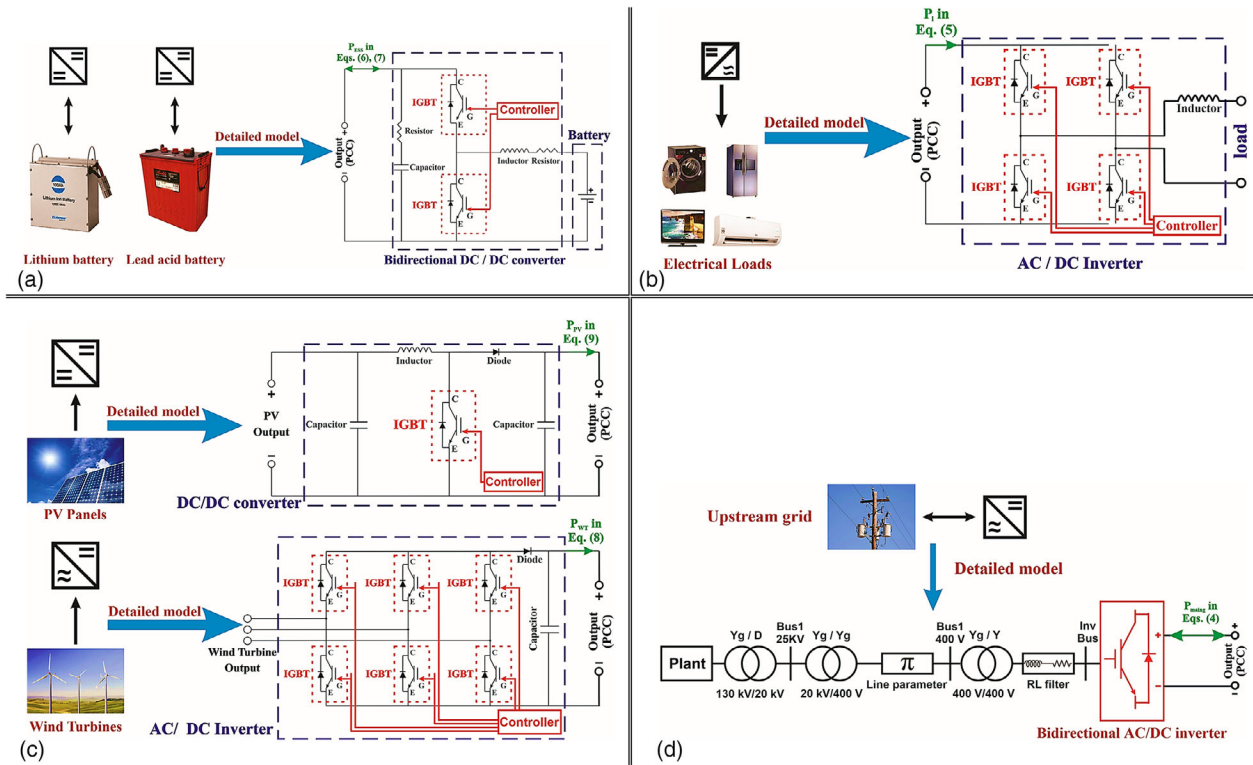


FIGURE 2 | The detailed modelling of the residential microgrid: (a) ESS, (b) electrical loads, (c) RESs and (d) upstream grid.

- RESs: PV panels inject power into the PCC via DC/DC converters, while WTs are connected through AC/DC inverters (Figure 2c).
- Upstream grid: The upstream grid is modelled with transformers and transmission line parameters and is connected to the

DC bus through a bidirectional AC/DC inverter, which allows the purchase and sale of electricity (Figure 2d).

The MPC-based EMS for decision-making integrates instantaneous microgrid data and predictive analytics with

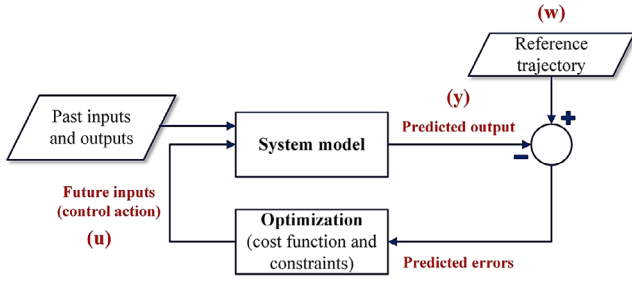


FIGURE 3 | The main structure of MPC [53].

performance-based prediction to optimise techno-economic operations in real time.

2.1 | Mathematical Formulation of Microgrid

The goal of this study is to optimise the economic and technical performance of a DC microgrid while adhering to operational constraints. The optimisation process employs the MPC method, which is well-suited for real-time energy management due to its ability to predict system behaviour and optimise control actions iteratively. The main structure of MPC is depicted in Figure 3.

The MPC framework uses a real-time dynamic model to predict the system's output based on historical and instantaneous measurements. Optimal control measures are computed iteratively at each time step to ensure adaptability to changing conditions. The quadratic equation of the MPC objective function is defined as:

$$J_{MPC} = \sum_{i=1}^{N_p} \| (y(t+i|t) - w(t+i))^2 \|_{N_p} + \sum_{i=1}^{N_c} \| (\Delta u(t+i-1))^2 \|_{N_c} \quad (1)$$

The first term minimises the error between the predicted system output (y) and the reference value (w) over the prediction horizon (N_p). The second term minimises the variations in control signals (u) over the control horizon (N_c) to ensure stability and smooth system operation [53]. For the DC microgrid, the power balance equation at any given time t is expressed as:

$$P_l(t) = P_{lith}(t) + P_{lead}(t) + P_{maing}(t) + P_{res}(t) \quad (2)$$

This equation ensures that the total power demand is always balanced by the combined contributions of the RES, ESS and the upstream grid, thereby maintaining supply-demand equilibrium in the microgrid. Also, The objective function aims to minimise the total cost of energy management while preventing excessive charging and discharging of the ESS. Consequently, the total objective function (J_{tot}) is the sum of the cost functions for individual components within the microgrid:

$$J_{tot} = J_{maing}(t+i) + J_l(t+i) + J_{lead}(t+i) + J_{lith}(t+i) + J_{RES}(t+i) \quad (3)$$

The parameters in Equation (3), J_{maing} , J_l , J_{lead} , J_{lith} and J_{RES} , are formulated based on Equation (1) to achieve optimal microgrid EMS.

Upstream grid: The objective function for power exchange with the upstream grid is defined to minimise energy costs, accounting for electricity price variations:

$$J_1 : J_{maing}(t+i) = \sum_{i=1}^{N_p} \left((P_{maing}(t+i))^2 \lambda(t+i) \right) + \sum_{i=1}^{N_c} \left(\Delta P_{maing}^2(t+i-1) \right) \quad (4)$$

where λ represents the electricity price at time t , which differs based on the tariff system. In each sample, based on the RES and ESS capacity and demand, with economic optimisation, the microgrid can purchase or sell electricity from or to the upstream grid at a price of λ .

Residential Loads: The objective function of residential loads is expressed to ensure the supply of demand as:

$$J_2 : J_l(t+i) = \sum_{i=1}^{N_p} \left((P_l(t+i) - P_d(t+i))^2 \right) + \sum_{i=1}^{N_c} \left(\Delta P_l^2(t+i-1) \right) \quad (5)$$

The first term of the function is related to changes in demand for the electric loads, which may differ due to the DR program during predetermined hours (Section 2.4). The second term is considered to ensure the supply of demand.

ESS: Separate the objective functions for the lead-acid and lithium-ion batteries are defined due to their unique characteristics to minimise deviations from reference SoC values, ESS depreciation and operational costs:

$$J_3 : J_{bat,lead}(t+i) = \left\{ \sum_{i=1}^{N_p} \left((SoC_{lead}(t+i) - SoC_{leadref})^2 \right) + \left\{ \sum_{i=1}^{N_c} (\Delta P_{lead}^2(t+i-1)) + \sum_{i=1}^{N_c} \frac{T \cdot CC_{b,lead} \cdot P_{lead}(t+i-1)}{C_{m,b,lead} \cdot Cy_{c,b,lead} \cdot V_{dc,bus} \cdot \eta_{b,lead}} \right\} \right\} \quad (6)$$

$$J_4 : J_{bat,lith}(t+i) = \left\{ \sum_{i=1}^{N_p} \left((SoC_{lith}(t+i) - SoC_{lithref})^2 \right) + \left\{ \sum_{i=1}^{N_c} (\Delta P_{lith}^2(t+i-1)) + \sum_{i=1}^{N_c} \frac{T \cdot CC_{b,lith} \cdot \eta_{b,lith} \cdot P_{lith}(t+i-1)}{2 C_{m,b,lith} \cdot Cy_{c,b,lith}} + Cost_{dep,lith} P_{lith}(t+i-1)^2 \right\} \right\} \quad (7)$$

The objective functions of lead and lithium batteries are represented in Equations (6) and (7), respectively. The first term is

TABLE 2 | Batteries parameters [54].

Parameters	Lead-acid battery	Lithium battery
Cyc	1500	3000
CC (\$/kWh)	1058.2	1659.9
C (Ah)	36700	10000
V _{dc_{bus}}	48	48
η	91.5%	93.75%
Cost _{dep} (\$/W ² h)	9×10^{-8}	—

related to the depreciation cost of the batteries to maintain ESS at optimal SoC, thereby preventing overuse of the ESS. The second term is associated with the operating cost of the batteries based on the battery characteristics, such as efficiency and number of cycles, which are determined in Table 2 by the manufacturers [54].

RES: The objective function of RES ensures their optimal utilisation, balancing generated and dispatched power. The uncertainty of RES including wind turbines and PV panels results from fluctuations in wind and solar radiation, which are affected by climate change. The power produced in a wind turbine based on wind speed is as follows [55]:

$$P_{WT}(t) = \begin{cases} 0, & v_w \langle v_{Cin} \text{ and } v_w \rangle v_{Cout} \\ P_w \left(\frac{v_w - v_{Cin}}{v_{wr} - v_{Cin}} \right), & v_{Cin} \leq v_w \leq v_{wr} \\ P_w, & v_{wr} \leq v_w \leq v_{Cout} \end{cases} \quad (8)$$

The PV power depends on environmental factors including solar radiation intensity, solar cell voltage and current based on temperature and fill-factor, which Equations (9)–(12) express the relevant equations [56].

$$P_{PV}(t) = A_{PV} \times FF \times V_i \times I_i \quad (9)$$

$$V_i = V_{oc} - K_{oc} T_c \quad (10)$$

$$I_i = G_i [I_{sc} + k_{sc} (T_c - T_{std})] \quad (11)$$

$$T_c = T_i + G_i \left(\frac{T_{nom} - 20}{0.8} \right) \quad (12)$$

Since the objective is to maximise the utilisation of RES when they are dispatched, the objective function of these resources is as follows, with binary variables α^{PV} and α^{WT} representing the presence of PV panels and wind turbines, respectively.

$$J_5 : J_{RES}(t+i) = \sum_{i=1}^{N_p} \alpha^{PV}(t) \left(\{ P_{PV}(t+i) - P_{PV}^{max}(t+i) \} \right)^2 + \sum_{i=1}^{N_p} \alpha^{WT}(t) \left(\{ P_{WT}(t+i) - P_{WT}^{max}(t+i) \} \right)^2, \\ P_{RES}(t) = P_{PV}(t) + P_{WT}(t) \quad (13)$$

2.2 | Operational Constraints

The operational constraints for the optimisation of microgrid components and efficient operation are as follows [57]:

$$SoC_{lead,min} \leq SoC_{lead}(t) \leq SoC_{lead,max} \quad (14)$$

$$SoC_{lith,min} \leq SoC_{lith}(t) \leq SoC_{lith,max} \quad (15)$$

$$P_{l,min} \leq P_l(t) \leq P_{l,max} \quad (16)$$

$$P_{maing,min} \leq P_{maing}(t) \leq P_{maing,max} \quad (17)$$

$$P_{lith,min} \leq P_{lith}(t) \leq P_{lith,max} \quad (18)$$

$$P_{lead,min} \leq P_{lead}(t) \leq P_{lead,max} \quad (19)$$

$$0 \leq P_{PV}(t) \leq P_{PV}^{max} \quad (20)$$

$$0 \leq P_{WT}(t) \leq P_{WT}^{max} \quad (21)$$

The conservative values are used for the SoC to prevent overcharging or undercharging in Equations (14) and (15), which increases longevity. The maximum and minimum consumption is determined in Equation (16), and the range of power exchanged with the upstream grid is based on the information of the power distribution company in Equation (17). Additionally, the maximum and minimum power of lithium and lead batteries, PV and WT are proportional to the capacity in Equations (18)–(21). The following constraints are considered to ensure a response from resources and to prevent sudden changes in power levels due to fluctuations in microgrid loads, distribution grid and ESS power [58].

$$\Delta P_{l,min} \leq \Delta P_l \leq \Delta P_{l,max} \quad (22)$$

$$\Delta P_{maing,min} \leq \Delta P_{maing} \leq \Delta P_{maing,max} \quad (23)$$

$$\Delta P_{lith,min} \leq \Delta P_{lith} \leq \Delta P_{lith,max} \quad (24)$$

$$\Delta P_{lead,min} \leq \Delta P_{lead} \leq \Delta P_{lead,max} \quad (25)$$

$$\Delta P_{RES,min} \leq \Delta P_{RES} \leq \Delta P_{RES,max} \quad (26)$$

2.3 | Load Curtailment

DR programs are used to optimally manage residential microgrids by reducing consumption during peak periods or high electricity prices. The utility company and the customers often agree on these periods ahead and only specific non-essential household appliances such as TVs and washing machines are turned off to ensure that daily activities are not interrupted [59]. During the load curtailment, consumption is calculated by

$$P_d(t) = \begin{cases} P_l(t) - P_{curt}(t), & \text{if } t_{start} \leq t \leq t_{end} \\ P_l(t) & , o.w. \end{cases} \quad (27)$$

P_d and P_l represent the demand and the total loads of the microgrid under normal conditions, respectively; P_{curt} is the total power of the loads participating in load curtailment between the start hour t_{start} and the end hour t_{end} .

2.4 | Objective Function Optimisation

The goal is to optimise the objective function, which is defined in Equation (3) while meeting the constraints Equations (14)–(26). Therefore, an optimisation problem is formulated using the MPC method and is solved by the MINLP algorithm as

$$\min_{\{P_{opt_maing}, P_{opt_lith}, P_{opt_lead}\}} J_{tot} = \sum_{k=1}^5 J_k \quad (28)$$

Constraints (14) to (26)

The nonlinearity in the optimisation model arises from the interaction between binary decision variables and continuous control variables, especially in Equation (28), where the charging/discharging state of ESS is formulated as a mixed integer decision problem. Since the objective function is obtained based on the MPC method in the form of a quadratic equation as expressed in Equation (1), it is required to optimise by MINLP and a linear formulation oversimplifies the problem and leads to suboptimal solutions.

2.5 | ESS Modelling Based on MPC

ESS is dynamically managed using an MPC framework to optimise SoC and minimise ESS depreciation while considering real-time changes in generation and demand. The ESS model in MPC is mathematically expressed as:

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) + B_d d(k) \\ x(k) &= SoC(k), u(k) = P_{ESS}(k), d(k) = P_{gen}(k) \\ &= P_{PV}(t) + P_{WT}(t) \end{aligned} \quad (29)$$

where A , B and B_d are system matrices of MPC-based modelling in Equation (29) that relate the ESS and the control variables. These matrices are derived from the ESS dynamics and battery characteristics:

$$A = 1, B = -\frac{\eta \cdot T}{C}, B_d = \frac{\eta \cdot T}{C} \quad (30)$$

where C represents the total capacity of the battery, T is the control time step and η is the efficiency factor of the battery. Therefore, by integrating Equations (29) and (30), a ESS model based on MPC is obtained, which, due to the unique characteristics of lithium and lead batteries, two separate models are expressed, respectively:

$$\begin{aligned} SoC_{opt_lith}(t+1) &= SoC_{lith}(t) - \frac{\eta_{lith} \cdot T}{C_{lith}} \{P_{opt_lith}(t) + (P_{dem}(t) - P_{lith}(t) \\ &\quad + P_{lead}(t) + P_{maing}(t) + P_{RES}(t))\} \end{aligned} \quad (31)$$

$$\begin{aligned} SoC_{opt_lead}(t+1) &= SoC_{lead}(t) - \frac{\eta_{lead} \cdot T}{C_{lead}} \{P_{opt_lead}(t) + (P_{dem}(t) - P_{lith}(t) \\ &\quad + P_{lead}(t) + P_{maing}(t) + P_{RES}(t))\} \end{aligned} \quad (32)$$

Consequently, the exchanged power with the upstream grid, lithium and lead battery power are calculated at each time interval. So, the exchanged power is determined by the calculated P_{opt_maing} and then the optimal SoCs are calculated. The values of the two variables P_{opt_lith} and P_{opt_lead} are placed in Equations (31) and (32), respectively and the optimal SoC value is calculated based on the instantaneous SoC (first term) in real time. The second term represents the changes in the SoC, which is obtained from the calculated optimal power of the batteries and considers the uncertainty related to RES and demand in MPC. In each battery, if the instantaneous value of the SoC is less than its optimal value, it means that the instantaneous SoC of the battery needs to be increased and the controller sends the charging command. Otherwise, the battery will be discharged.

3 | Microgrid Energy Management System

The MINLP method is employed in this study to optimise the microgrid energy management system. MINLP is an optimisation technique that handles problems involving both discrete decision-making variables, such as charging/discharging of ESS and continuous variables, like power flows. This capability makes it particularly suitable for addressing the nonlinearities and diverse operational constraints inherent in real-world microgrid applications [60, 61].

The objective function in Equation (28) is optimised by MINLP to determine three key outputs: Exchanged power with the upstream grid, P_{opt_maing} , lithium battery power, P_{opt_lith} and lead battery power, P_{opt_lead} . The proposed framework is shown in Figure 4, and the EMS takes three primary input variables: the demand (P_{dem}) the power produced by RESs (P_{RES}) and the reference SoC of the batteries ($SoC_{lithref}$ and $SoC_{leadref}$).

The parameters in objective functions are derived from instantaneous measurements and microgrid simulations over a 24 h horizon. The predictive data feeds into the MINLP framework, which solves the objective function and calculates the optimal exchanged power between the microgrid and the upstream grid (P_{opt_maing}) and optimal power of batteries (P_{opt_lith} and P_{opt_lead}), then calculates the corresponding optimal SoC values (SoC_{opt_lith} and SoC_{opt_lead}) through Equations (31) and (32), respectively. The integration of uncertainties in RES production and demand fluctuations into the MPC framework ensures dynamic responses to operational variability, significantly enhancing the system's real-time adaptability. Figure 5 presents the block diagram of the proposed techno-economic optimisation framework and the EMS iteratively executes the following steps to optimise energy management over a 24 h horizon:

- Step one (initialisation): Define initial parameters for optimal power of ESS and exchanged power between the microgrid and the upstream grid based on their initial capacity.

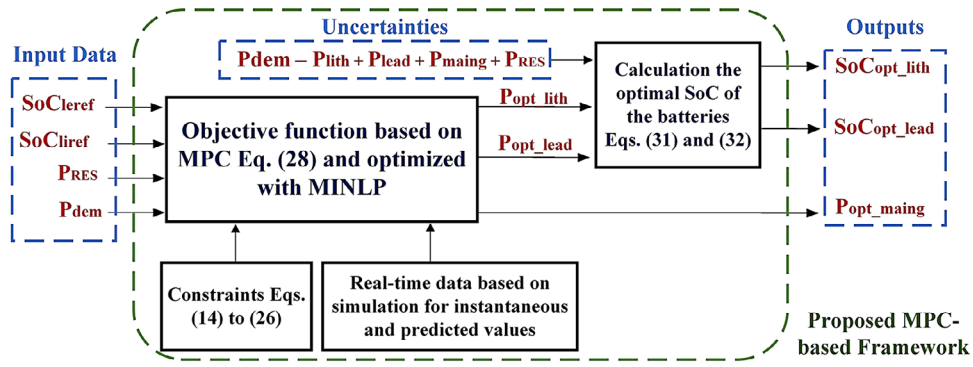


FIGURE 4 | The structure of the proposed framework.

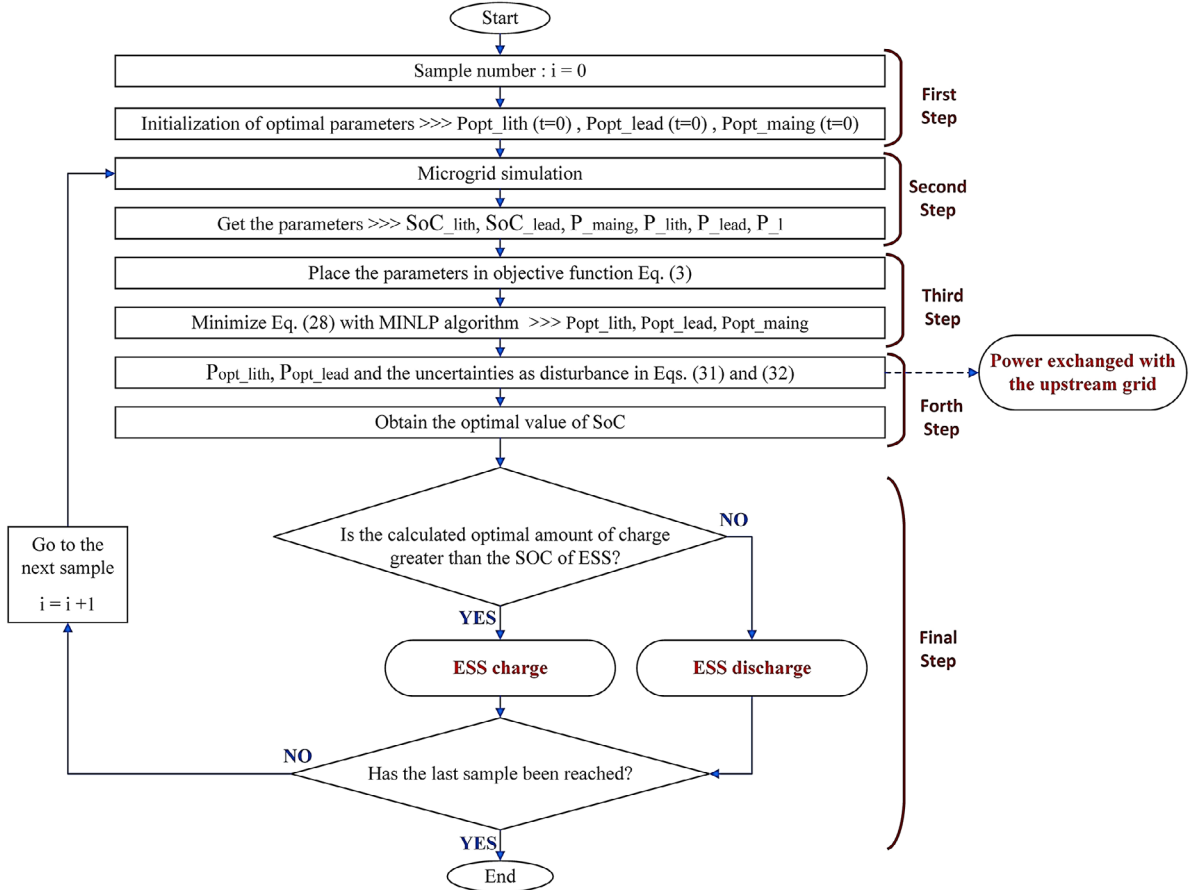


FIGURE 5 | The proposed framework algorithm.

- Step two (data collection): Perform a 24 h microgrid simulation to provide predictive data for the proposed framework, incorporating instantaneous measurements.
- Step three (optimisation): Solve the objective function using the MINLP method with performance-based prediction data which are obtained in the previous step, while satisfying operational constraints such as SoC, exchanged power and RES availability to optimise the power exchanged with the upstream grid.
- Step four (SoC adjustment): Calculate the optimal SoC values for both batteries (SoC_{opt_lith} and SoC_{opt_lead}) by integrating

RES production uncertainty, instantaneous SoC values and real-time performance-based predictive data.

- Final step (control commands): Compare the optimal and instantaneous SoC values to issue charging or discharging commands for the ESS operation.

This iterative process allows the EMS to manage both discrete decisions, such as ESS charge/discharge modes and continuous variables, such as optimal power flow, using real-time performance-based predictive data. As a result, it increases the scalability of the system by managing complex interactions

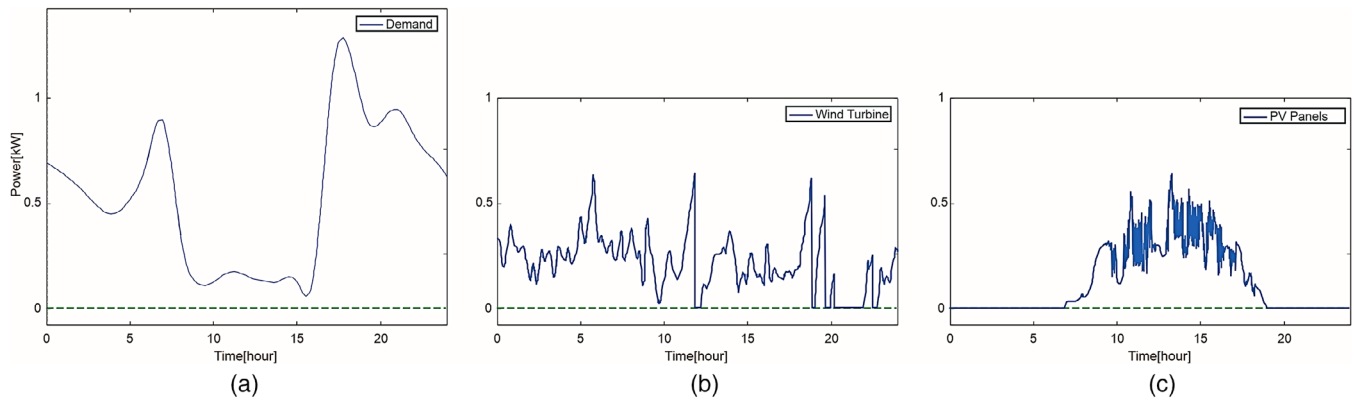


FIGURE 6 | Electricity consumption and RES production curve: (a) demand curve, (b) WT and (c) PV [62].

between microgrid components and the integration of the formulation and modelling of microgrid components based on MPC offers several advantages in the proposed method:

- **Real-time adaptability:** The EMS directly incorporates RES uncertainties and demand fluctuations into the ESS management. Because, the operation of the microgrid in each sample is simulated for 24 h based on instantaneous conditions and these data are used as performance-based predictive data instead of predefined data. So, any changes in components and microgrid operation are considered in EMS.
- **Longevity optimisation of ESS:** By minimising SoC changes, the framework reduces ESS depreciation, thereby extending the longevity of lithium and lead-acid batteries. This technical goal ensures sustainable microgrid operation.
- **Economic efficiency:** The EMS optimises power exchanges with the upstream grid based on dynamic electricity prices, including real-time, three-tariff and fixed-tariff schemes. This economic goal results in significant cost savings.
- **Scalability:** This framework allows seamless adaptation to larger or more complex microgrids with diverse energy sources and ESS configurations. As a result, other or greater sources can be integrated into the MINLP framework by adjusting the input variables and constraints.

4 | Simulation Results

The proposed framework has been evaluated under diverse scenarios to validate its technical adaptability and economic efficiency in integrating RES and ESS. Figure 6(a)–(c) depict the demand and RES production curves, which are derived from residential microgrid datasets to ensure realistic simulation conditions over a 24 h horizon [62]. If the 24 h optimisation is performed with a shorter window, reliable results cannot be obtained due to uncertainties in renewable resources and demand. On the other hand, with a longer optimisation horizon, the computational overhead increases significantly without a fundamental change in the results. Therefore, a 24 h window is a balanced trade-off that captures the daily electricity price changes and renewable generation while maintaining a manageable computational load.

TABLE 3 | The DC microgrid components and specifications [62].

Microgrid component	Value
Maximum upstream grid power	600 (kW)
Maximum load	150 (kW)
Maximum PV panel capacity	75 (kW)
Maximum wind turbine capacity	70 (kW)
Lead battery	48 (Vdc), 1500 (Ah)
Lithium battery	48 (Vdc), 2000 (Ah)

The PV generation profile exhibits variability caused by partial shadowing and fluctuations in solar irradiance. Similarly, the wind turbine output, although generally less variable than PV, experiences pronounced fluctuations due to wind speed variations, including periods of cut-off during low or high wind speeds. In contrast, the residential demand profile displays smoother patterns, as its high sampling rate reduces short-term variations in consumption patterns. These realistic profiles ensure that the simulation captures the operational challenges faced by microgrids in real-world scenarios. Moreover, the inclusion of these variations allows the EMS to demonstrate its ability to dynamically manage the variability of RES production while ensuring demand and supply. The specification of the microgrid components are presented in Table 3. To evaluate the adaptability under diverse economic conditions, three types of electricity pricing are considered: real-time tariffs, three-tariff pricing and fixed tariffs, as shown in Figure 7. These pricing mechanisms simulate energy market conditions, which enable the EMS to optimise power flows and minimise costs effectively under varying economic constraints [63]. A load curtailment strategy is implemented during peak-demand hours (17:00–21:00) to reduce grid stress, as shown in Figure 8, which ensures better resource utilisation and minimises operational costs.

Key microgrid constraints, such as SoC, exchanged power with the upstream grid and RES output, are summarised in Table 4. These constraints balance computational efficiency with operational requirements. The simulation also incorporates variations

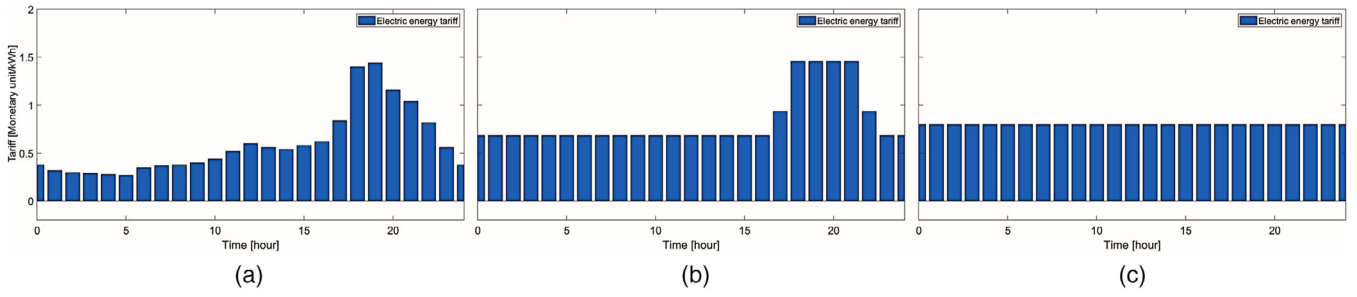


FIGURE 7 | The electricity pricing tariff [63]: (a) real-time tariffs, (b) three-tariff pricing and (c) fixed tariffs.

TABLE 4 | Maximum and minimum bounds of the microgrid constraints [62].

Parameters	Minimum bound	Maximum bound
Power of the loads	$P_{l, \min} = 60$ (kW)	$P_{l, \max} = 150$ (kW)
Power exchanged with the upstream grid	$P_{\text{maing}, \min} = -600$ (kW)	$P_{\text{maing}, \max} = 600$ (kW)
Power of the PV and WT	$P_{\text{WT}, \min} = P_{\text{PV}, \min} = 0$ (kW)	$P_{\text{WT}, \max} = 75$ (kW), $P_{\text{PV}, \max} = 60$ (kW)
Power of the lead and lithium battery	$P_{\text{lead}, \min} = -96$ (kW), $P_{\text{lith}, \min} = -72$ (kW)	$P_{\text{lead}, \max} = 96$ (kW), $P_{\text{lith}, \max} = 72$ (kW)
Variation in power of loads and	$\Delta P_{l, \min} = 2$ (kW)	$\Delta P_{l, \max} = 20$ (kW)
Variation in power of the upstream grid and RES	$\Delta P_{\text{maing}, \min} = \Delta P_{\text{RES}, \min} = -10$ (kW)	$\Delta P_{\text{maing}, \max} = \Delta P_{\text{RES}, \max} = 10$ (kW)
Variation in power of the lead and lithium battery	$\Delta P_{\text{lith}, \min} = \Delta P_{\text{lead}, \min} = -10$ (kW)	$\Delta P_{\text{lith}, \max} = \Delta P_{\text{lead}, \max} = 10$ (kW)
SoC of the lead and lithium battery	$\text{SoC}_{\text{lead}, \min} = 35\%$, $\text{SoC}_{\text{lith}, \min} = 30\%$	$\text{SoC}_{\text{lead}, \max} = 75\%$, $\text{SoC}_{\text{lith}, \max} = 85\%$

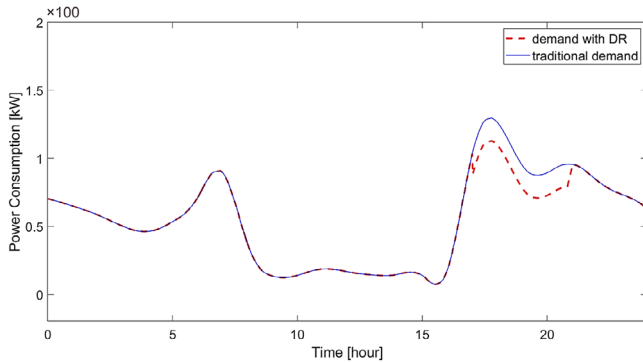


FIGURE 8 | Demand curve with load curtailment (DR program).

in RES production and load demand to provide the EMS with performance-based predictive data to ensure robust decision-making. To evaluate the performance of the proposed EMS, three scenarios are investigated, the reference SoC for lithium and lead batteries ($\text{SoC}_{\text{lith,ref}}$ and $\text{SoC}_{\text{lead,ref}}$) is assumed to be 60% in all scenarios. Also, MPC-required parameters including sample time, prediction and control horizon are chosen to balance computational efficiency with real-time adaptability [62].

$$N_p = 6 \text{ (sample)}, N_c = 2 \text{ (sample)}, T = 30 \text{ (s)} \quad (33)$$

The computational burden of the proposed method depends on the number of decision variables, including prediction horizon

(N_p) and control horizon (N_c) and the number of binary variables RESs in the microgrid. Thus, it is essential to choose appropriate values for N_p and N_c to avoid over-calculations. Because the proposed method is compared with an MPC-based method [62] and to ensure a fair comparison, these MPC parameters are the same. This guarantees that any observed performance improvements arise from the methodological advancements of the proposed framework rather than differences in parameter selection. However, reducing the prediction and control horizons results in sudden and undesirable fluctuations in the results. This occurs because a shorter horizon limits the optimisation's ability to account for rapid fluctuations in renewable generation and electricity prices. Increasing these parameter values significantly increases computational volume in both methods. However, no meaningful improvements in microgrid performance are observed and the results remain almost the same.

4.1 | Scenario (I): Reference Scenario Using the Conventional MPC-Based Approach

In the first scenario, an MPC-based approach is applied, as described in [62], which focuses on optimising energy resources in a residential microgrid while integrating a demand response program (Figure 8). This method evaluates the power-produced prediction based on predetermined data and compares it with the demand, then sends charge and discharge commands to the ESS. It lacks adaptability to dynamic operational constraints and fails

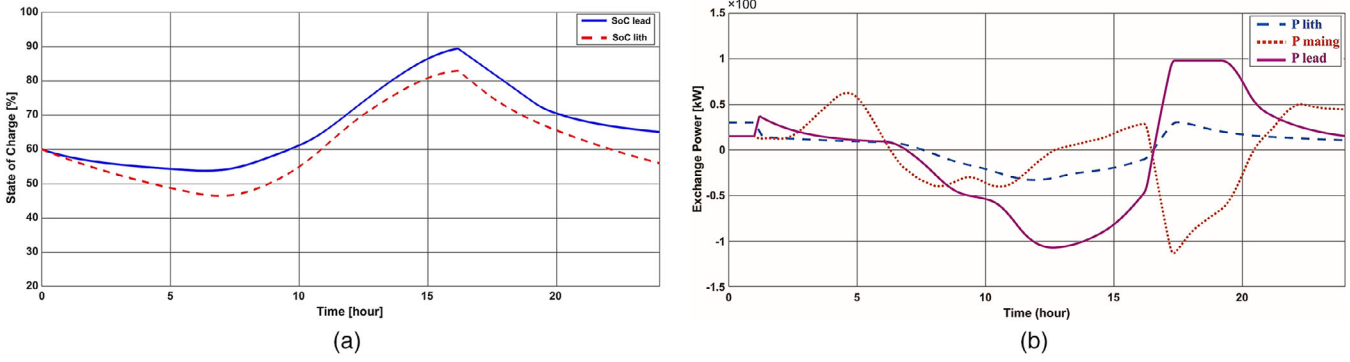


FIGURE 9 | The reference scenario using the conventional MPC-based approach with PV: (a) SoC, (b) power flow.

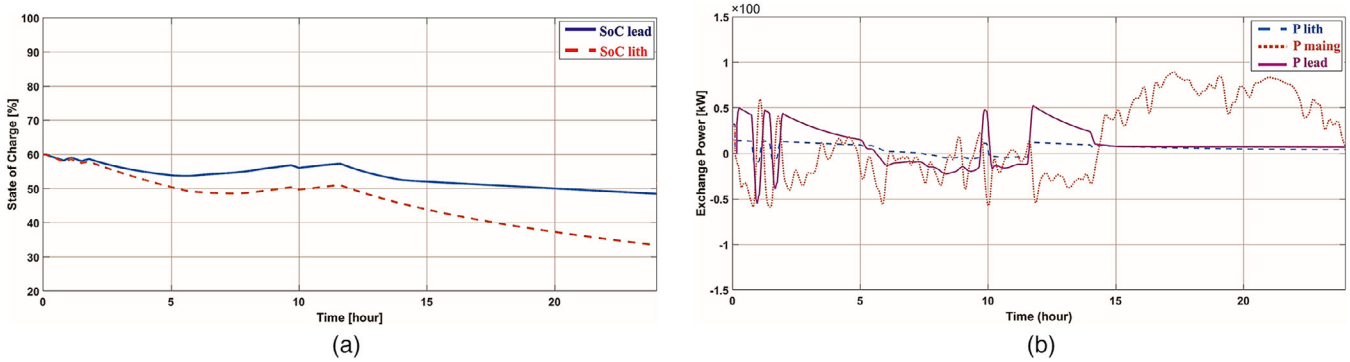


FIGURE 10 | The reference scenario using the conventional MPC-based approach with WT: (a) SoC and (b) power flow.

to incorporate economic and technical factors simultaneously, such as economic saving and ESS depreciation, respectively. This method is chosen to ensure meaningful comparison within the same MPC-based control framework, rather than comparing fundamentally different energy management techniques or other types of MPC approaches like distributed MPC [64], Stochastic MPC [65], etc. Also, to ensure a proper comparison between methods, all key parameters related to MPC, including control horizons, prediction horizons and sample time have been considered the same.

This scenario is simulated under two configurations: (i) PV panels and (ii) WT as the RES. In the first case, only PV is considered as the RES (Figure 6c). The results in Figure 9 highlight significant SoC changes, particularly during PV production hours (7:00 to 19:00). In the early hours, the ESS undergoes significant discharging and since the ESS capacity is insufficient, the microgrid relies on purchased power from the grid. In contrast, during hours of high PV production, the ESS charge leads to increased SoC and surplus power is sold to the upstream grid. The controller neglects operational constraints, which leads to excessive use of ESS, resulting in large SoC changes and increased ESS depreciation. Also, economic factors, such as electricity price, are ignored, which causes the overuse of stored power during low-price hours and the purchase of more power during high-price periods, leading to suboptimal economic performance in the conventional approach.

In the second case, wind turbines (WT) are considered (Figure 6b), and the SoC and power flow results are depicted in

Figures 10a and b, respectively. The intermittent nature of wind power generation results in lower average power production compared to the previous case. Consequently, the ESS discharges more frequently, which results in a lower SoC on reference (60%) and more ESS depreciation (Figure 10a). The lack of consideration for electricity price signals leads to increased power purchases from the upstream grid during peak-price hours (17:00 to 24:00), as shown in Figure 10b. Uneconomical operation arises from excessive grid reliance during peak periods and insufficient utilisation of cheaper RES power for ESS charging.

4.2 | Scenario (II): the Proposed MPC-Based Framework

In this scenario, the proposed MPC-based framework and the DR program (Figure 8) are considered. Initially, the microgrid is simulated with only PV panels as the RES (Figure 6c). The proposed framework optimizes energy management in the microgrid while adhering to operational constraints specified in Equations (14) to (26).

The results in Figure 11a demonstrate that the proposed framework effectively reduces the SoC changes, which keeps it close to the reference value of 60%. During PV production hours (7:00 to 19:00), the SoC change remains minimal, with a maximum deviation of approximately 5% from the reference value. During non-production hours (20:00 to 7:00), the maximum SoC deviation increases to around 15%, still significantly lower than

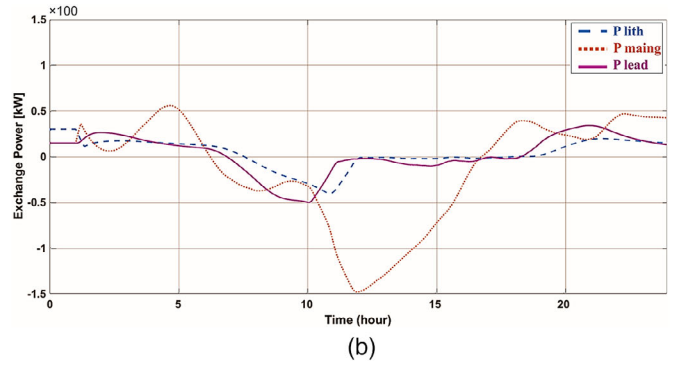
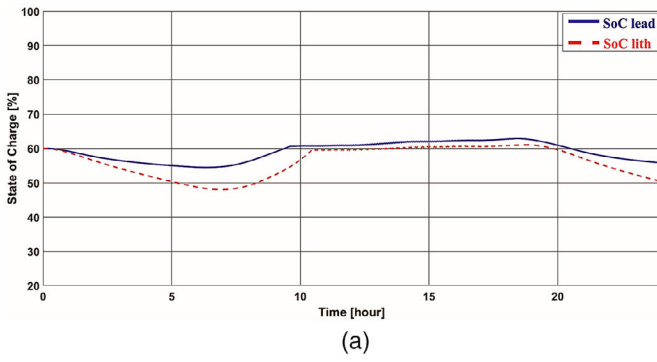


FIGURE 11 | The proposed framework with PV: (a) SoC and (b) power flow.

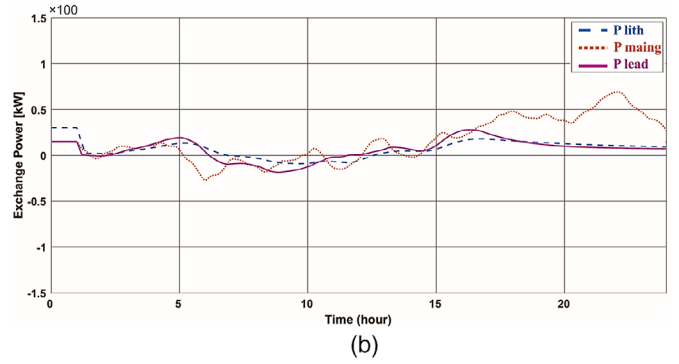
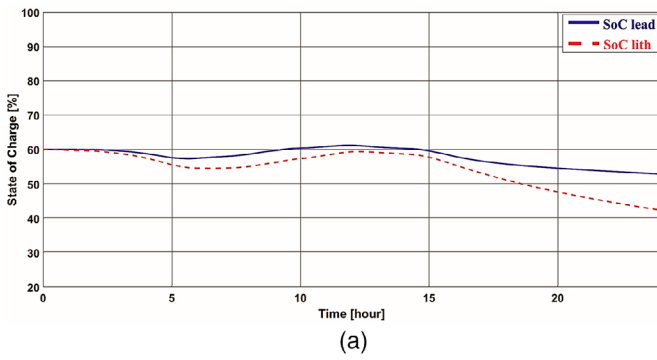


FIGURE 12 | The proposed framework WT: (a) SoC and (b) power flow.

the reference scenario. This reduced SoC change minimises ESS depreciation to ensure prolonged ESS longevity. Furthermore, the surplus power generated by PV during production hours is sold to the upstream grid and the framework reduces purchased power from the grid during peak-price periods (17:00 to 24:00), as illustrated in Figure 11b. Compared to the reference scenario, the proposed framework demonstrates a more economical operation by prioritising power exchange decisions based on electricity price.

The system's performance is further evaluated by replacing wind turbines, which introduces lower power generation. Figure 12a shows that the SoC remains close to the reference value (60%) due to the efficient use of WT power for charging and discharging during 24 h and reduced SoC changes, extending ESS longevity. Additionally, Figure 12b highlights the system's economic benefits, as the microgrid minimises purchased power from the upstream grid during peak-price hours. During the evening (20:00 to 24:00), when electricity prices are highest, the proposed framework optimally discharges the ESS to supply demand to ensure minimal reliance on expensive grid power. While this strategy results in a slightly larger SoC deviation from the reference value towards the end of the day, it aligns with the economic optimisation goals.

The results emphasise the effectiveness of the proposed framework in seamlessly integrating RES, ESS, and DR programmes for microgrid management. Compared to the reference scenario, the framework significantly reduces SoC changes, leading to a considerable decrease in ESS depreciation, thus, optimising

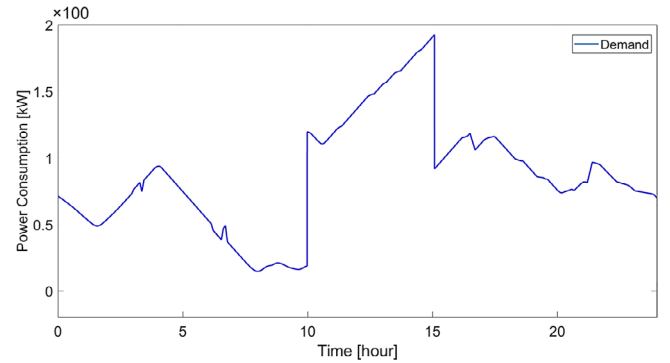


FIGURE 13 | Demand increase in scenario III.

power flow and ESS operations achieves economic and technical goals. To further demonstrate the framework's adaptability, the third scenario evaluates its performance under sudden increases in demand and added fluctuations, simulating unpredictable conditions.

4.3 | Scenario (III): The Proposed MPC-Based Framework for Large-Scale Microgrid

In this scenario, a sudden increase of 100 kW in the microgrid load is introduced between 10:00 and 15:00 with fluctuations that are close to the peak load of the network during the day (150 kW), as shown in Figure 13. This change leads to an increase in demand during these hours to nearly three times the initial

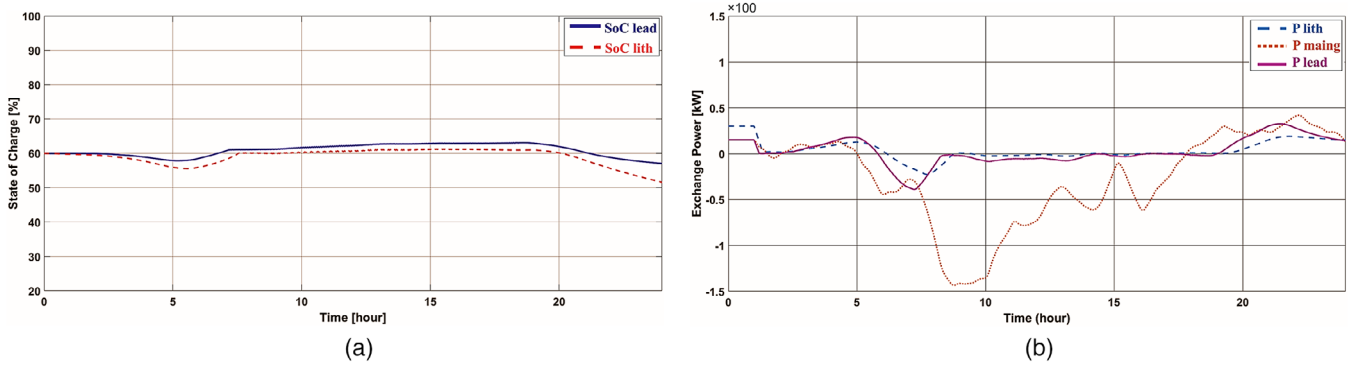


FIGURE 14 | The proposed framework in demand increase in scenario III: (a) SoC and (b) power flow.

TABLE 5 | Results of purchasing and selling electricity with real-time tariff.

		Case studies					
		Reference scenario			Proposed framework		
		PV	Wind turbines	Demand increase	PV	Wind turbines	Demand increase
Economic performance	Purchased energy cost [¥]	103,311	111,041	132,599	90,201	97,225	116,704
	Sold energy income [¥]	18,352	12,514	9,824	25,225	20,306	15,695
	Total cost paid [¥]	84,959	98,527	122,775	64,976	76,919	101,009

value. Both PV panels and wind turbines are used as RESs to manage the load and the proposed EMS effectively optimises ESS operations, maintaining SoC levels close to the reference value (Figure 14a), which reduces ESS depreciation. Even under fluctuating demand conditions, the controller avoids excessive ESS discharging or charging, preventing depreciation. Also, the proposed framework demonstrates economic performance by minimising dependency on the upstream grid, especially during peak hours (Figure 14b). This scenario effectively represents a large-scale microgrid with more dynamic loads and higher renewable energy capacity.

4.4 | Discussion: Economic Performance of the Proposed Framework

In all scenarios, real-time tariff is considered to determine the electricity price and three-tariff and fixed tariff pricing are also simulated in this study. The results are the same in the conventional MPC-based approach due to ignoring the electricity price, the proposed framework has a slight difference in the results of SoC and power flow, but the difference in microgrid costs is significant. Therefore, the economic performance of the framework is presented in Tables 5 to 7 under real-time tariffs, three-tariff pricing and fixed tariffs, respectively, which demonstrates the superior cost-effectiveness of the proposed method across all scenarios. Among these tariffs, the real-time tariff system achieves the most significant cost savings, with reductions of approximately 20% compared to the reference scenario. This is attributed to the dynamic nature of real-time pricing, which aligns power exchanges with hourly price fluctuations and allows the system to minimise costs by optimally exchanging power.

The proposed MPC-based energy management system reduces operational costs by dynamically optimising purchased power, RESs and ESS in real-time-based electricity prices. Moreover, by minimising ESS depreciation and integrating demand response strategies, the framework ensures long-term economic sustainability. In the reference scenario, the conventional MPC-based method neglects techno-economic optimisation simultaneously and uses predefined prediction data, which results in a consistent SoC and power flow across all tariff systems and leads to suboptimal energy management. Conversely, the proposed framework integrates instantaneous electricity prices into the optimisation framework (Equation 4), dynamically adjusting the power exchange based on hourly price fluctuations. As shown in Tables 5–7, the proposed framework achieves notable cost savings in all scenarios:

- For PV-only systems, the total cost paid is reduced by approximately 23.52%, 21.84% and 16.73% compared to the reference scenario in real-time tariffs, three-tariff pricing and fixed tariffs, respectively.
- For systems with wind turbines, the total cost paid decreases by approximately 21.93%, 17.99% and 17.20%.
- In the case of increased demand and RESs, the proposed framework still outperforms the reference scenario, reducing costs by 17.72%, 15.25% and 15.82%.

As shown in Table 8, the proposed EMS not only delivers substantial economic savings but also ensures reduced maximum SoC changes, which highlights its capability to minimise ESS depreciation. Notably, the most significant cost reductions occur under real-time tariff pricing, where the method's real-time

TABLE 6 | Results of purchasing and selling electricity with three-tariff pricing.

		Case studies					
		Reference scenario			Proposed framework		
		PV	Wind turbines	Demand increase	PV	Wind turbines	Demand increase
Economic performance	Purchased energy cost [¥]	108,524	123,300	144,664	96,674	106,913	127,394
	Sold energy income [¥]	13,054	16,914	11,554	22,064	19,670	14,585
	Total cost paid [¥]	95,470	106,386	133,110	74,610	87,243	112,809

TABLE 7 | Results of purchasing and selling electricity with a fixed tariff.

		Case studies					
		Reference scenario			Proposed framework		
		PV	Wind turbines	Demand increase	PV	Wind turbines	Demand increase
Economic performance	Purchased energy cost [¥]	105,702	116,546	146,785	93,759	99,533	128,924
	Sold energy income [¥]	18,555	15,101	10,512	21,197	15,541	14,216
	Total cost paid [¥]	87,147	101,445	136,273	72,562	83,992	114,708

TABLE 8 | Techno-economic performance of proposed framework compared to conventional method.

	Electricity prices tariff system								
	Real-time tariff			Three-tariff pricing			Fixed tariff		
	Case study								
	PV	Wind turbines	Demand and RESs increase	PV	Wind turbines	Demand and RESs increase	PV	Wind turbines	Demand and RESs increase
Technical performance: maximum SoC changes [%]	15.4%	18%	10.2%	16.3%	17.4%	10.5%	16.4%	18.2%	13.5%
Economic performance: economic savings [%]	23.52%	21.93%	17.72%	21.84%	17.99%	15.25%	16.73%	17.20%	15.82%

adaptability to dynamic price signals yields the greatest economic benefits. This performance validates the proposed MPC-based framework as an advanced solution to optimise both technical and economic objectives in residential microgrids.

Considering photovoltaic panels and wind turbines separately in the performance analysis of the proposed method demonstrates energy management under high-penetration renewable energy generation and its variability. Also, the third scenario effectively represents a large-scale microgrid with more dynamic loads and increased renewable energy capacity. The results demonstrate that the proposed method successfully manages the additional load while maintaining power balance to ensure minimal fluctuations in the SoC of the batteries. The maximum

SoC deviation remained below 13.5%, and the economic saving was more than 15.2%, confirming the stability of the method under higher capacity conditions, as shown in Table 8. The findings presented in this study are sufficient to demonstrate its applicability to larger microgrids. Furthermore, the proposed MPC-based energy management framework is designed to be adaptable to different microgrid configurations, including grid-connected and islanded modes. While this study focuses on grid-connected operation, the same optimisation principles can be applied to islanded microgrids with modifications to demand-side management. The approach can also be extended to multi-microgrid systems, enabling cooperative power sharing between interconnected microgrids for enhancing power system resilience against high-impact events.

The computational efficiency of the proposed framework is considered to ensure real-time feasibility. By formulating the problem as an MINLP optimisation with a rolling horizon approach, the framework maintains a balance between solution accuracy and computational speed. The prediction and control horizons are selected to prevent excessive computation times while preserving system operational and economic efficiency. The simulation results confirm that execution times remain within real-time constraints, making the method practical for DC residential microgrid applications. Although the proposed approach uses performance-based prediction from real-time system performance, future enhancements could incorporate machine learning and adaptive control techniques to further improve its performance. Machine learning-based forecasting models could enhance prediction accuracy, while adaptive MPC could dynamically adjust control parameters based on system conditions. The accuracy of the MPC forecast performance can be improved for initial demand data (Figure 6a), renewable energy generation (Figures 6b and c) and electricity price (Figure 7) in future research. Also, using adaptive MPC to tune control and prediction horizons based on real-time operating conditions might improve system robustness in response to unexpected conditions.

5 | Conclusion

This study presents an optimal EMS designed for residential microgrids that integrate RES and ESS to address the technical and economic challenges posed by rising electricity demand and RESs. The proposed framework based on MPC incorporates DR programmes and utilises instantaneous and performance-based prediction data derived from dynamic simulations. This adaptive approach enables precise, dynamic and efficient energy management under changes in operational conditions, unlike conventional methods which rely on static and predefined datasets. The MPC-based objective function is optimised using the MINLP method, integrating ESS and RES with the upstream grid. This optimisation process is implemented using Simulink/Matlab software, which is a powerful and widely used platform in the field of energy systems optimisation, which can handle complex and large-scale problems. The performance of the proposed framework is evaluated across three scenarios: a reference case using a conventional MPC-based approach, the implementation of the proposed framework, and a scenario involving sudden demand increases and RES fluctuations. The results underscore the framework's ability to achieve substantial technical and economic improvements. By incorporating essential operational constraints, such as electricity price, component constraints and power exchanges with the upstream grid, the framework has the following key benefits:

- **Enhanced ESS longevity:** The proposed MPC-based EMS significantly reduces ESS depreciation by minimising SoC changes, with a maximum variation of 23.52% among all scenarios and conditions, which ensures prolonged ESS longevity and enhances the economic feasibility of residential microgrids.
- **Economic efficiency:** By leveraging real-time pricing signals and optimising power exchanges, the framework achieves

considerable cost savings, particularly under real-time tariff systems. Optimal utilisation of surplus power further maximises financial benefits for residential consumers.

- **Adaptability to real-time optimisation:** The integration of real-time optimisation and performance-based prediction data ensures that the EMS adapts effectively to RES variability and demand fluctuations, maintaining reliable and cost-effective microgrid operation even under challenging conditions.

The proposed framework demonstrates high potential for practical deployment in emerging high-technology residential microgrids. These technologies enable the framework to dynamically manage energy resources, reduce consumer costs and alleviate power system stress during peak demand periods. Furthermore, the scalability and adaptability of the framework make it suitable for diverse microgrid configurations, enabling efficient operation across various scenarios and complexities. This framework can be extended to support decentralised energy trading among prosumers in a multi-microgrid. For larger microgrid networks, an MPC-based approach can be explored to enable multi-agent coordination among interconnected microgrids. This condition can enhance power system resilience against high-impact events. By combining real-time optimisation, enhanced ESS longevity and economic efficiency, this study establishes a robust and scalable EMS solution for residential microgrids. Future research opportunities include extending the proposed framework to commercial and industrial microgrids with their unique consumption patterns and operational demands. Moreover, incorporating advanced demand management strategies, such as load shifting and integrating emerging distributed energy resources (DERs) like fuel cells and electric vehicles in hybrid AC/DC microgrids could further evaluate the framework's adaptability and performance. For residential-scale applications, embedded systems such as Raspberry Pi, NVIDIA Jetson, or programmable logic controllers (PLCs) or real-time units (RTUs) in industrial microgrids can be used to implement the optimisation algorithm using smart meters and ESS management systems to ensure real-time decision-making and system stability. These advancements would support the seamless integration of microgrids into broader power systems, addressing the evolving complexities of global energy needs.

Description of Parameters

Variables

λ	Instantaneous electricity price (α)
SoC_{lead}	Lead battery SoC (%)
SoC_{lith}	Lithium battery SoC (%)
$SoC_{leadref}$	Reference value of lead battery SoC (%)
$SoC_{lithref}$	Reference value of lithium battery SoC (%)
I_b	Battery current (A)
t_{start}	Load curtailment start time (h)
t_{end}	Load curtailment end time (h)
y	System output signal

w	Reference value of the output signal
u	Control measure signal
J_{MPC}	Objective function based on the MPC
J_{tot}	Objective function of microgrid
J_{maing}	Objective function of the upstream grid
J_{lead}	Objective function of lead battery
J_{lith}	Objective function of lithium battery
J_l	Objective function of electricity load
v_w	Rated wind speed (m/S)
v_{wr}	Wind speed (m/S)
v_{cin}, v_{cout}	Cut-in and Cut-out wind speed (m/S)
C	Battery capacity (kWh)
A_{PV}	Area of the PV panel (m ²)
FF	Fill factor of the PV panel
V_i, I_i	Voltage and current of the solar cell (V, A)
G_i	Solar irradiance (W/m ²)
V_{oc}, I_{sc}	Open-circuit voltage and short-circuit current of PV (V, A)
K_{oc}, K_{sc}	Temperature coefficients of the open-circuit voltage and short-circuit current of PV (V/°C)
T_c, T_{std}	Instantaneous and standard temperature for PV (°C)
x, u, d	State variable, control variable, and disturbance in MPC
α^{PV}, α^{WT}	Binary variables indicating the presence of PV and WT

Electrical Power Variables (KW)

P_{maing}	Upstream grid
P_{lith}	Lithium battery
P_{lead}	Lead battery
P_l	Electrical loads
P_d	Microgrid demand
P_{RES}	Generation by renewable energies
P_{curt}	Load curtailment
P_{PV}	Power generated by PV panels
P_{WT}	Power generated by Wind Turbines
$\Delta P_{maing}, \Delta P_l, \Delta P_{lith}, \Delta P_{lead}, \Delta P_{RES}$	The difference in exchanged power with the upstream grid, loads, lithium, lead batteries, and RES
P_{opt_maing}	Optimal exchanged power with the upstream grid
P_{opt_lith}	Optimal power of lithium battery
P_{opt_lead}	Optimal power of lead-acid battery

Constant Values

$C_{m,b}$	Maximum battery capacity (kWh)
CC_b	Battery acquisition cost (\$/kWh)
Cyc_b	Number of life cycles of the battery
$V_{dc_{bus}}$	Direct current bus voltage (V)
η_{lead}	Lead battery efficiency (%)
η_{lith}	Lithium battery efficiency (%)
$Cost_{dep}$	Battery depreciation cost (\$/W ² h)
N_p	Prediction horizon in MPC
N_c	Control horizon in MPC
T	Sampling time (s)
i	Sample number
T_{nom}	The nominal operating cell temperature (°C)
P_w	Rated power of wind turbine (kW)
A, B, B_d	System matrices for ESS model in MPC

Author Contributions

Sina Roudnil: writing – original draft, visualization, validation, software, resources, project administration, methodology, investigation, formal analysis, data curation, conceptualization. **Saeid Ghassem Zadeh:** writing – review & editing, validation, supervision, project administration, investigation, funding acquisition, conceptualization. **Mohammad Reza Feyzi:** writing – review & editing, supervision. **Amir Aminzadeh Ghavifekr:** writing – review & editing, methodology, formal analysis.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable. No new data is generated and the article describes entirely theoretical research.

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